

Quantum Computing

Part 2 - Hardware by Modality: Trapped Ion Qubits

Compiled by the Spaced Research Ledger

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A trapped-ion quantum computer uses individual charged atoms, held in place by electric fields inside a vacuum chamber, as its qubits. Every ion of a given element is identical, which removes the manufacturing variation that troubles the chips in Part 1. Ions also hold their state far longer, and any ion in a chain can interact with any other rather than only its neighbours.

The cost is speed. Operations take microseconds where superconducting gates take nanoseconds, roughly a thousandfold difference. The field's central engineering question is whether that penalty can be reduced faster than superconducting machines can improve their fidelity.

A second theme runs through this part. Most trapped-ion systems control their ions with lasers, which are bulky and hard to scale. Several groups are replacing lasers with electronics or microwaves, and Sections 1, 4 and 5 are all, in part, about that shift.

Section 1 covers IonQ, Section 2 Quantinuum, Section 3 AQT, Section 4 eleQtron, and Section 5 the University of Oxford.

1. IonQ

IonQ's recent work is about removing lasers from the control path. Its electronic control approach produced the highest two-qubit gate fidelity in this part, and did it without the elaborate cooling that such results normally require.

Recent Developments

IonQ demonstrated electronically controlled two-qubit gates with an estimated error of $8.4(7) \times 10^{-5}$, which is a fidelity above 99.99% [1]. The result did not require ground-state cooling. Error stayed at or below 5×10^{-4} even with average phonon occupation as high as 9.4(3) on the gate mode.

That second sentence carries the significance. Ground-state cooling means chilling an ion's motion to its lowest possible level before operating on it, and it is slow. A gate that tolerates a warm, vibrating ion removes a step from every operation.

Current State

IonQ's architecture gives every qubit in a single ion chain access to every other, through a shared motional bus [2]. For electronically controlled gates between distant qubits, ions must first be shuttled to the same physical location, a principle the company compares to charge-coupled devices.

The commercial machines are smaller than that headline result suggests. IonQ's Forte and Forte Enterprise processors each hold 36 qubits, built from trapped ytterbium ions in a compact surface linear Paul trap and controlled by 355-nanometre laser transitions [3]. Median two-qubit fidelities measured 99.3% on Forte and 99.5% on Forte Enterprise, with single-qubit rotations around 99.98%.

An independent benchmark supports the connectivity claim. A peer-reviewed paper in the journal Quantum tested a Forte system configured as a single 30-qubit chain and benchmarked all 435 possible qubit pairs [4]. Testing every pair rather than a sample is what distinguishes a genuine all-to-all claim from an average.

The electronic control technology replaces laser-based control with precision electronics manufactured on standard semiconductor chips, putting the control components directly onto microfabricated chips [5].

2. Quantinuum

Quantinuum builds the highest-performing trapped-ion machines in this part and physically moves its ions around the chip to do it. Its current system is also the hardware behind the logical-qubit results in Part 6.

Recent Developments

Quantinuum's Helios system is a 98-qubit processor using barium hyperfine qubits, with all-to-all connectivity provided by a rotatable ion storage ring and a four-way junction [6]. Average two-qubit gate infidelity is $7.9(2) \times 10^{-4}$, which is about 99.921% fidelity. Single-qubit infidelity is $2.5(1) \times 10^{-5}$, about 99.9975%. State preparation and measurement infidelity is $3.3(5) \times 10^{-4}$.

Current State

Helios uses a quantum charge-coupled device architecture, usually shortened to QCCD, which physically transports ions within the trap rather than relying only on optical addressing [5]. The architecture organizes ions into separate zones for computation, storage and readout, and shuttles them between zones as needed [9]. That keeps interaction zones small and high-fidelity while the total qubit count grows on a single chip.

The previous generation set the standing benchmark. Quantinuum's H2 system, with 56 all-to-all connected qubits, reached a quantum volume of 2^{25} in September 2025 [5]. Quantum volume is a single number combining qubit count, fidelity and connectivity. That value is the highest reported for any machine in this collection.

H2's component error rates are documented in detail. It has a state-preparation-and-measurement error rate of 0.15%, a two-qubit gate error rate of 0.14(1)%, long-range connectivity, and mid-circuit measurement and reset crosstalk errors at or below 2×10^{-5} [7]. Mid-circuit measurement means reading one qubit while others keep computing, which error correction depends on.

Two benchmarking figures round out the picture. The 20-qubit H1-1 system reported median single-qubit fidelity of 99.996% and median two-qubit fidelity of 99.9% [8]. The 56-qubit H2 device showed 99.997% and 99.87% for the same two measures [8].

3. AQT

AQT's contribution is not peak performance but form factor. Its machines run at room temperature in standard equipment racks, which changes where a quantum computer can be installed.

Recent Developments

AQT's LYNX series reached a quantum volume of 32,768 with all-to-all connectivity [10]. The system fits a 19-inch rack-mounted enclosure and operates at room temperature.

Current State

AQT builds compact, rack-mountable trapped-ion systems using calcium ions [11]. Its PINE system fits into two 19-inch racks, supports up to 50 qubits, and runs at room temperature drawing under 2 kilowatts. Compare that with the dilution refrigerators the superconducting systems in Part 1 require. A machine that runs on rack power in a normal room has a different installation cost than one that does not.

The company was spun out of the University of Innsbruck by Rainer Blatt, Peter Zoller and Thomas Monz, continuing that university's calcium-ion research line [11].

4. eleQtron

eleQtron replaces lasers with magnetic gradients and microwave fields. The approach is less proven than the alternatives in this part, and several of its strongest claims come from the company and its funding consortium rather than from independent publication.

Current State

The core technology is called magnetic gradient induced coupling, and it uses microwave fields rather than lasers to control trapped-ion qubits [14].

The strongest published result is a speed improvement. A Physical Review X paper from June 2025, co-authored by a researcher now at eleQtron, demonstrated a laser-free radio-frequency two-qubit entangling gate [12]. It ran roughly an order of magnitude faster than the previous best radio-frequency gate, which took 740 microseconds. The same work increased qubit coherence time by three orders of magnitude by decoupling from field noise.

That gate reached an experimental entanglement negativity of 0.48, with an uncertainty range from plus 0.02 to minus 0.06 [13]. The maximum possible value is 0.5, so the result sits close to a maximally entangled two-qubit state. It used only a single continuous radio-frequency field per qubit, which matters for scaling.

The broader performance claims are weaker evidence. eleQtron's EU project documentation reports record-low crosstalk between qubits, high single- and two-qubit gate fidelities, and coherence times up to 10 seconds [14]. It also claims all-to-all connectivity and operation at higher temperatures than superconducting systems. These are company and consortium statements rather than independently published results, and should be read as such.

5. University of Oxford

Oxford holds the reference results in this field, both the laser-driven benchmark that stood for years and the all-electronic work that displaced it. Its spinout was acquired by IonQ, which connects this section back to Section 1.

Recent Developments

Oxford demonstrated microwave-driven two-qubit gates on calcium hyperfine qubits in a cryogenic surface trap [15]. Two operating points were reported: a 154-microsecond gate with 1.0(2)% error, and a slower 331-microsecond gate with 0.5(1)% error. The trade between speed and accuracy is visible directly in those two numbers.

Current State

The single-qubit result is the strongest in this collection. An Oxford-based team demonstrated all-electronic control of trapped-ion qubits in a microfabricated seven-zone trap, reaching an average single-qubit gate fidelity of at least 99.99912(8)% [16]. It worked by switching interaction fields on and off between zones, and was published in PRX Quantum in October 2025.

The history explains why the recent results matter. Oxford's 2016 laser-driven two-qubit gate reached 99.9(1)% fidelity, with single-qubit fidelity of 99.9934(3)%, on calcium hyperfine qubits [17]. That was the standing reference point for trapped-ion gate fidelity for years.

It has since been passed twice. Oxford Ionics, an Oxford spinout now part of IonQ, demonstrated 99.97% two-qubit fidelity on a calcium system using electronic qubit control [11]. IonQ's own result on ytterbium hardware, in Section 1, then exceeded that.

About This Report

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